Susceptibility of Adult *Aethina tumida* (Coleoptera: Nitidulidae) to Entomopathogenic Fungi

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ABSTRACT *Aethina tumida* Murray (Coleoptera: Nitidulidae) is an invasive parasite species in populations of honey bees, *Apis mellifera* L. Aiming toward substitution of chemical control, we here identified a naturally occurring fungal pathogen of adult *A. tumida* from its endemic range in South Africa [*Metarhizium anisopliae* (Metschnikoff) Sorokin variety *anisopliae* strain FI-203]. The susceptibility of adult beetles ($n = 400$) to this fungus and to three other generalist entomopathogenic fungal isolates [*Metarhizium anisopliae*, *Beauveria bassiana* (Balsamo) Vuillemin, and *Hirsutella illustris* Minter & Brady] was assessed using spore suspension bioassays. The data revealed significantly increased mortality in the *B. bassiana* (74.00 ± 8.94%) and *M. anisopliae* variety *anisopliae* (28.00 ± 16.43%) tests but not in the *H. illustris* (2.00 ± 4.47%) and *M. anisopliae* (12.00 ± 8.37%) groups. The results indicate a potential for entomopathogenic fungi as an alternative control of *A. tumida*.

KEY WORDS *Aethina tumida*, *Apis mellifera*, biological control, entomopathogenic fungi

Aethina tumida Murray (Coleoptera: Nitidulidae) is a parasite of honey bee, *Apis mellifera* L., colonies native to sub-Saharan Africa (Hepburn and Radloff 1998) and has recently become an invasive species (Hood 2004, Neumann and Elzen 2004). Although African honey bee subspecies are relatively tolerant *A. tumida* infestations, European-derived honey bees are considerable less tolerant (Hood 2004, Neumann and Elzen 2004). The rapid spread and high reproductive potential of *A. tumida* (Neumann et al. 2001, Muerrle and Neumann 2004), both within colonies as well as in stored hive products (Ellis et al. 2002, Ellis 2004), together with its capacity to hibernate in honey bee clusters makes it a potential worldwide pest of apiculture (Neumann and Elzen 2004).

So far, control of *A. tumida* has focused on chemical treatments (Baxter et al. 1999a, b; Elzen et al. 1999), bearing the risks of both pest resistance and residues in the hive products (Neumann and Elzen 2004). Biological control, using microbial pathogens and in particular entomopathogenic fungi, has the potential as an alternative to chemical insecticides (Lacey et al. 2001). Indeed, fungal pathogens are often highly host-specific and nontoxic to vertebrates (Lacey et al. 2001), suggesting that they also might constitute a feasible alternative control for *A. tumida.*

A. tumida belongs to the coleopteran family Nitidulidae with members mainly being phytophagous, fungivorous, or saprophytic on fermenting fruit or plant matter (Scholtz and Holm 1996). Some Nitidulidae are known to vector a variety of disease-causing microorganisms, for example, spores of *Aspergillus flavus* Link:Gray (Vega et al. 1995, Wicklow 1995). The fungivorous beetle *Carpophilus freemani* Dobson was shown to be able to transport conidia of *Beauveria bassiana* (Balsamo) Vuillemin via fecal material in laboratory tests, which makes it an efficient candidate as a natural vector of *B. bassiana* (Bruck and Lewis 2002). *A. tumida* was frequently reported to promote fermentation of honey within combs and may therefore be a vector for fungal microorganisms (Lundie 1940, Schmolke 1974, Swart et al. 2001). *A. tumida* feed on fruit (Ellis et al. 2002) and decaying or fermenting hive products and reproduce readily on old and moldy combs (Muerrle and Neumann, unpublished data). These observations suggest that *A. tumida* may be tolerant to a variety of microbial pathogens, which naturally occur in their environment. However, Lundie (1940) first reported a potential unidentified fungal control agent, when noticing high mortality of adult beetles during laboratory rearing. Similarly, Ellis et al. (2004), noted increased larval and pupal mortality of *A. tumida* and suggested that two entomopathogenic fungi, *A. flavus* or *Aspergillus niger* van Tieghem might be responsible. Mortality of adult *A. tumida* caused by an unidentified fungus was also observed during beetle mass rearing (Muerrle and Neumann 2004) for experimental purposes (T.M.M. and P.N., unpublished data).

The fungal genera *Beauveria*, *Metarhizium*, and *Hirsutella* are generalist entomopathogens with spe-

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cies- and strain-dependent differences in specificity and virulence against a range of insect groups (Samson et al.1988, Kendrick 1992, Lacey et al. 2001). They have a cosmopolitan distribution and can be isolated from insects, mites, soil, and a variety of other substrates (Boucias and Pendland 1998). Some strains are commercially available and successfully used against a range of insect or mite pests (Kendrick 1992, Douthwaite et al. 2000, Lacey et al. 2001). Therefore, *Beauveria, Metarhizium,* and *Hirsutella* seem to be candidates for the control of *A. tumida.*

Here, we report the results of the isolation of the unidentified fungal pathogen observed during a previous beetle mass-rearing program (Muerrle and Neumann 2004) and its identification by using molecular techniques. Furthermore, we evaluated the susceptibility of adult *A. tumida* to this isolated fungus and to three additional isolates of other entomopathogenic fungi (*B. bassiana, H. illustris,* and *M. anisopliae*) indigenous to South Africa.

Materials and Methods

Laboratory Rearing. Adult *A. tumida* ($n = 25$) were collected from three different beehives from an apiary in Stones Hill (33° 20′ S, 26° 34′ E), South Africa, in June 2004. A laboratory culture was established according to routine protocols (Neumann et al. 2001, Muerrle and Neumann 2004). All adults were introduced into a plastic rearing box (30 by 40 by 30 cm) together with three combs containing brood, pollen, and honey, where oviposition and larval development took place. Mature wandering larvae were collected after 14 d and introduced into plastic pupation jars (∅ 7 cm; height 14 cm) filled with moist, pasteurized soil. After eclosion, the adult beetles were stored in smaller plastic containers (15 by 20 by 8 cm). Food was provided ad libitum in the form of honey-soaked cotton wool.

Fungal Strain, Isolation, Cultivation, and Morphological Identification. One unidentified fungal species, which was obtained from a deceased adult *A. tumida* during a previous beetle mass-rearing program (>36,000 adults; Muerrle and Neumann 2004) was isolated. Isolation of the fungal strain was conducted after surface sterilization of the *A. tumida* specimen (immersion in 50% ethanol for 5 s). Then, inoculation of the whole specimen was performed on malt extract agar (MEA) (3% malt extract, 0.5% soya peptone, and 1.5% agar), in a sterile plastic petri dish (∅ 9 cm) at 28°C in a constant environment room. Preliminary morphological identification of the isolate to genus was carried out by microscopic examination of hyphal and conidial material by using the tape mount method (Samson et al. 1988).

Molecular Identification. *DNA Extraction.* The fungal isolate was inoculated into nutrient broth (0.1% malt extract, 0.2% yeast extract, 0.5% peptone, and 0.8% sodium chloride) and cultivated for 7 d at 28°C, on a shaker at 130 rpm. Genomic DNA was extracted from 40 ml of this culture. The mycelial material was pelleted by centrifuging at 2500 rpm for 5 min and manually crushed in 2% hexadecyltrimethylammonium bromide. After addition of proteinase K, the preparation was incubated at 37°C for 1 h and then at 65°C for 15min. All subsequent extraction steps were carried out according to standard protocols (Rose et al. 1990). Integrity of the genomic DNA was assessed by agarose gel electrophoresis.

Internal Transcribed Spacer (ITS) Region Polymerase Chain Reaction (PCR). The ITS region of the rRNA gene was amplified using the *Taq*PCR Mastermix kit (QIAGEN, Valencia, CA) and universal primers ITS1-F (5′-CTT GGT CAT TTA GAG GAA GTA A-3′) and ITS4 (5′-TCC TCC GCT TAT TGA TAT GC-3′) after Henrion et al. (1994) with the following modifications. The PCR started with denaturation at 94°C for 2 min, followed by 30 cycles of 94°C for 30 s, 45°C for 45 s, and 72°C for 60 s and ended with final elongation at 72°C for 7 min. Fragment size was assessed by agarose gel electrophoresis.

DNA Sequencing. The PCR product was purified with a Wizard SV gel and PCR clean-up system (Promega, Madison, WI). Purified PCR products were sequenced using an ABI Prism BigDye Terminator v. 3.1 and the chemicals and protocols of the supplier (Applied Biosystems, Foster City, CA). Identical primers as mentioned above were used for sequencing of the ITS-1 locus. A BLAST search on the sequence data and their subsequent submission to GenBank (http://www.ncbi.nlm.nih.gov/) for registration was carried out.

Further Fungal Isolates. Three further isolates, indigenous to South Africa, included one isolate of *Metarhizium anisopliae* (Metschnikoff) Sorokin PPRI 6717 (isolated from termites), one isolate of *Beauveria bassiana* (Balsamo) Vuillemin PPRI 6706 (isolated from termites), and one isolate of *Hirsutella illustris* Minter & Brady PPRI 6407 (isolated from aphids). These were obtained from the National Collection of Fungi, Plant Protection Research Institute, Tshwane (=Pretoria), South Africa. All four isolates were each inoculated onto MEA ($n = 10$ plates each), for proliferation and spore production and were kept in darkness at 28°C.

Preparation of Conidial Suspensions. Conidia from eight 2-wk-old cultures of each isolate were harvested by flooding the plates with sterile distilled water (sdH_2O) and carefully stirring to break up conidial chains and to suspend the spores. The concentrated spore suspensions from eight plates of the same cultures were mixed and diluted with sdH_2O to 250 ml. All spore suspensions were stored in darkness at 4°C. Viability of the spores was assessed by two germination tests on MEA. The first test was carried out on the day of preparation of the suspensions, the second test 25 d later, after termination of the experiments. For each isolate, 5 μl of suspension were inoculated onto MEA (n = two plates each) and fungal growth was observed for 7 d.

Virulence Tests. All tests ($n = 40$) were carried out on treatments and controls of equal size ($n = 10$ *A. tumida* in five replicates for each isolate). Adult *A. tumida* were randomly chosen from ≈2,500 individuals and introduced into petri dishes (∅ 9 cm). All petri dishes contained four layers of circular filter paper (∅ 5 cm), cotton wool (∅ 3 cm), and a circular food container (∅ 1.5 cm; height 0.5 cm) with sugar patty (icing sugar/honey, 5:1). All ingredients were autoclaved before introduction. In the treatments, 10 ml of spore suspension was applied using a sterile pipette. The controls were given an equal amount of sdH_2O. The spore suspensions or the sdH_2O were only applied to the cotton wool before the introduction of the live specimen. All treatments and controls were kept in darkness at 18–25°C. The mortality of *A. tumida* was recorded daily for 24 d.

Re-isolation of Pathogens. Koch's postulate was applied to confirm infectivity of the initial fungal isolates (Madigan et al. 2003). Re-isolation of pathogens was achieved by using bodies of dead *A. tumida* from treatments with obvious signs of fungal infection (hyphal outgrowth) as inocula on MEA, after 5 s of surface sterilization by immersion in 50% ethanol. Re-isolates were kept in darkness at 28°C for 1 wk. Identification of the re-isolates was carried out by microscopic examination and visual comparison with the initial cultures (see above).

Statistical Analyses. Fisher's exact and chi-square tests were used to test for differences in mortality between treatments and controls of *A. tumida.* We tested for differences in mortality between isolates by using an analysis of variance (ANOVA) and post hoc comparisons. All tests were performed using Statistica (StatSoft, Tulsa, OK).

Results

Fungal Strain Identification and Registration. Microscopic examination of the unidentified fungal isolate resulted in preliminary identification as a *Metarhizium* sp. The BLAST search of the ITS1 region revealed the following match (forward: e-value, −125; identity, 95%; backward: e-value, −107; identity, 89%) to a sequence registered under the accession number AF136376: *M. anisopliae* variety *anisopliae* strain FI-203. The submitted sequence data were registered and provided with the accession numbers: DQ062084 and DQ062085, which were released in June 2005 (GenBank). In addition, a sample of the fungal isolate was submitted to the National Collection of Fungi, Plant Protection Research Institute, Tshwane, where it was deposited under the registration number PPRI 7842.

Preparation of Conidial Suspensions. Viability of spores could be confirmed in all germination tests of conidial suspensions, with the exception of the spore suspension *H. illustris* in the second viability test (after termination of the experiments).

Virulence Tests. *M. anisopliae* variety *anisopliae.* The mortality of adult *A. tumida* was significantly higher in the treatments than in the controls (χ^2 = 8.58, df = 1, P = 0.003; Fisher's exact test, P = 0.006). From day 7 onward, a slight increase in mortality was observed in the treatments. Mortality increased slowly until the end of the experiment on day 24, where it reached an average of 28.00 ± 16.43%. The mortality in the control groups remained at 6.00 ± 13.42% after day 7 (Fig. 1).

M. anisopliae. No significant differences in the mortality of adult *A. tumida* were found between the treatments and the controls (χ^2 = 3.84, df = 1, P = 0.05; Fisher's exact test, P = 0.112). A slight increase of mortality in the treatment groups was observed from day 16 onward, with an increasing trend until the end of the experiment on day 24, where it reached an average of 12.00 ± 8.37%. Mortality of the control groups remained at 2.00 ± 4.47% after day 7 (Fig. 1).

B. bassiana. The mortality of adult *A. tumida* was significantly higher in the treatments than in the controls (χ^2 = 55.01, df = 1, $P < 0.0001$; Fisher's exact test, $P < 0.0001$). From day 7 onward, a rapid increase of mortality in the treatment groups was found. Mortality increased quickly with a slightly weaker trend at the end of the experiment, where it reached an average of 74.00 ± 8.94%. Mortality of the control groups remained at 2.00 ± 4.47% after day 5 (Fig. 1).

H. illustris. No significant differences in the mortality of adult *A. tumida* were found between the treatments and the controls (χ^2 = 0.00, df = 1, P = 1.0; Fisher's exact test, $P > 0.05$). No increase of mortality in the treatments and control groups was observed during the entire experiment. Mortality of treatment and control groups remained at 2.00 ± 4.47% for both groups from day 10 onward.

Comparisons between Isolates. We found highly significant differences between the four isolates (ANOVA: F = 46.121, df = 3, $P < 0.0001$). *M. anisopliae* variety *anisopliae* was significantly different from *B. bassiana* (Tukey's honestly significant difference [HSD], $P < 0.05$) and *H. illustris* (Tukey's HSD, $P < 0.05$) but not from *M. anisopliae* (Tukey's HSD, $P > 0.05$). *M. anisopliae* was significantly different from *B. bassiana* (Tukey's HSD, $P < 0.05$) but not from *H. illustris* (Tukey's HSD, $P > 0.05$). Finally, *B. bassiana* increased mortality significantly more compared with all other groups (Tukey's HSD, $P < 0.05$ in all cases).

Re-isolation of Pathogens. Fungal pathogens were re-isolated from seven deceased specimens of treatment groups *M. anisopliae* variety *anisopliae* and from eight deceased specimens of treatment groups *B. bassiana.* The initial pathogens were detected on MEA after 7 d of incubation in all cases for *M. anisopliae* variety *anisopliae* and in seven cases for *B. bassiana.* Fungal and bacterial contaminations were detected on four positive re-isolates of *M. anisopliae* variety *anisopliae* and on all positive and the negative re-isolate of *B. bassiana.* Re-isolation of fungal pathogens from deceased specimens of treatment groups *M. anisopliae* and *H. illustris* were not carried out, because no significant differences were found between treatments and controls.

Discussion

For the first time, we were able to identify a naturally occurring fungal pathogen of adult *A. tumida* from its endemic range in South Africa. The genetic

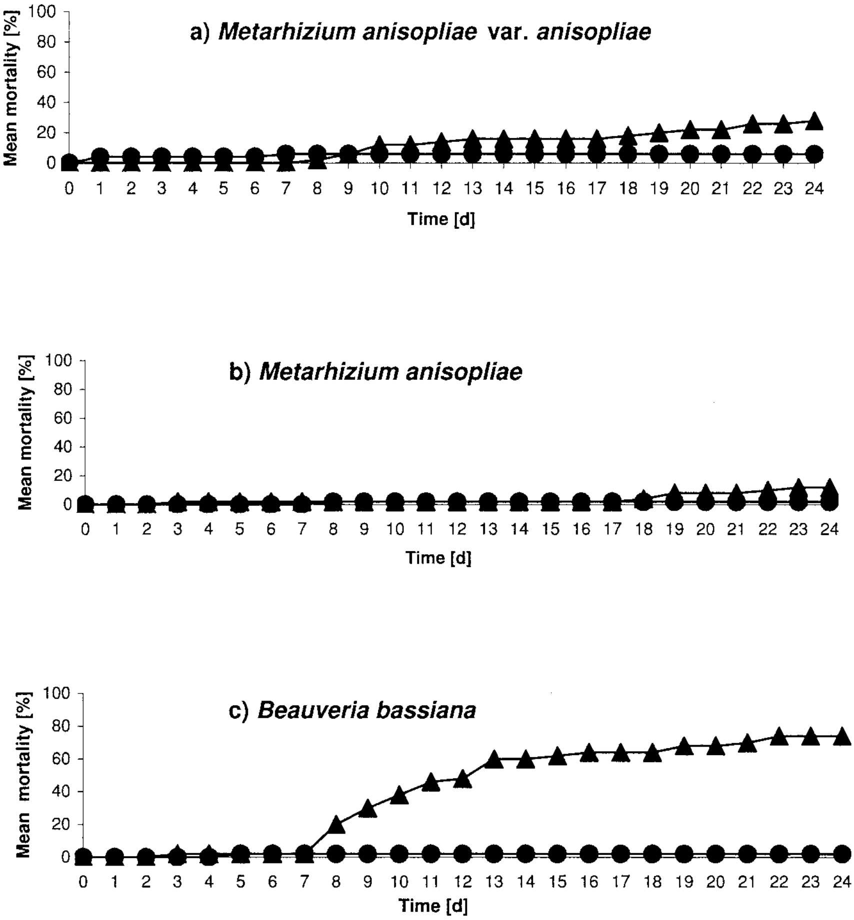

Fig. 1. The mortality (%) of adult *A. tumida* after inoculation with *M. anisopliae* variety *anisopliae* (a), *M. anisopliae* (b), or *B. bassiana* spore suspension (c) (▲, treatments; ●, controls).

data showed that this fungal pathogen was most closely related to *M. anisopliae* variety *anisopliae* strain FI-203, but nevertheless we found 11% sequence divergence for the ITS region. Our data also show considerable susceptibility of adult *A. tumida* to the entomopathogenic fungi *B. bassiana.* Re-isolation of these two fungal pathogens confirmed the infection with the initial pathogen and thereby Koch's postulate. *H. illustris* and *M. anisopliae* had no significant effect. Furthermore, *B. bassiana* increased the mortality significantly more than all other isolates. This supports the notion of species- and strain-dependent differences in specificity and virulence of entomopathogenic fungi (Samson et al. 1988, Kendrick 1992, Lacey et al. 2001). Although exposed to a variety of microbial pathogens in their natural environment, the relatively high susceptibility of adult *A. tumida* to entomopathogenic fungi is remarkable. Preliminary tests on *A. tumida* larvae indicate a lower susceptibility to the tested fungi (data not shown), confirming studies that fungal infectivity in insects is often closely linked to specific developmental stages because of changes in cuticular composition and environmental conditions (Samson et al. 1988, Harris et al. 2000, James et al. 2003). Given that *A. tumida* larvae are actually less susceptible, a successful control of adults would nevertheless significantly lower losses caused by larval feeding as well. Fungal control of coleopteran pests of stored grain for example similarly relies mainly on increased adult beetle mortality (Adane et al. 1996, Moino et al. 1998, Rice and Cogburn 1999, Kassa et al. 2002).

It has recently been suggested that the fungi *A. flavus* and *A. niger* may cause mortality of larvae and pupae of *A. tumida* (Ellis et al. 2004). However, these two fungi are known to cause mortality of larval (=stonebrood) and adult honey bees (Morse and Flottum 1997, Swart et al. 2001) and serious diseases in humans (Bennett and Klich 2003). This makes them rather unsuitable as biological control agents against *A. tumida.* The lack of toxicity of *B. bassiana* and *M. anisopliae* to mammals and humans (Langewald et al. 1997; USEPA 1999, 2003) renders those isolates as potential candidates for biological control. However, nontarget effects of these fungi were recorded for other insects (Samson et al. 1988, Danfa and van der Valk 1999, Ivie et al. 2002, Humber and Hansen 2004), including honey bees (Shaw et al. 2002). However, nontarget effects may be desirable, for example, to the mite *Varroa destructor* Anderson & Trueman, which is susceptible to *Hirsutella, Beauveria,* and *Metarhizium* species (Kanga et al. 2002, Peng et al. 2002, Shaw et al. 2002). Susceptibility of honey bees to *B. bassiana* and *M. anisopliae* strains was only evaluated in laboratory experiments (Shaw et al. 2002), which may not necessarily reflect field conditions. Thus, before recommending *B. bassiana* and *M. anisopliae* strains as a potential combined biological control for *A. tumida* and *V. destructor* in honey bee colonies, further field testing is needed to limit nontarget effects on honey bees and wild insect populations.

Field surveys of *A. tumida* in its endemic range in Africa, followed by inoculation of appropriate media from collected samples, also may provide an additional direct way to screen for new control agents, e.g., entomopathogenic fungi that have some beetle virulence but are present in otherwise healthy honey bee colonies. Indeed, the isolated fungal pathogen *M. anisopliae* variety *anisopliae* strain FI-203 strain cultured from an endemic *A. tumida* population was significantly more virulent than another strain of the same species [*Metarhizium anisopliae* (Metschnikoff) Sorokin PPRI 6717], which was isolated from termites. Thus, surveys in the endemic range of *A. tumida* seem to be a key point in developing a species-specific efficient agent. However, the tested strain of *B. bassiana* seemed most virulent, although it also was isolated from termites. This supports the notion that a whole range of factors determines susceptibility of a species to entomopathogenic fungi (Lacey et al. 2001). Both *B. bassiana* and *M. anisopliae* have wide host ranges, including major insect orders (Humber and Hansen 2004). However, it is now widely recognized that these two species contain a diverse assemblage of genotypes and probably make up "species complexes," with individual isolates exhibiting highly restricted host ranges and isolates recovered from a target host generally being more virulent than isolates from nonrelated species (Inglis et al. 2001). We therefore suggest that screening for more fungal isolates from *A. tumida* and testing different modes of simple application, which may be part of an integrated pest management, is essential. Because it is plausible to assume that there is variation in response to entomopathogenic fungi across the naturally occurring populations of *A. tumida* in Africa, susceptibility also should be investigated for a range of beetle origins in future studies.

We conclude that the results are promising, especially with regard to an ongoing search for more specific and more virulent strains of fungal pathogens and the possibilities for the successful development of an efficient mycoinsecticide against *A. tumida.*

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